Representative Spectral Correlation Network for Multisource Remote Sensing Image Classification

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***Abstract*—Hyperspectral image (HSI) and synthetic aperture radar (SAR)/light detection and ranging (LiDAR) data offer complementary spectral and structural information for land-cover classification. However, their effective fusion remains challenging due to two major limitations: the spectral redundancy in high-dimensional HSI and the heterogeneous characteristics between multisource data. To this end, we propose a representative spectral correlation network (RSCNet), a novel multisource image classification framework specifically designed to address the above challenges through spectral selection and adaptive interaction. The network incorporates two key components: 1) *key band selection module* (KBSM) that adaptively selects task-relevant spectral bands from the original HSI under cross-source guidance, thereby alleviating redundancy and mitigating information loss from conventional principal component analysis (PCA)-based spectral reduction. Moreover, the learned band subset exhibits highly discriminative spectral structures that align with discriminative semantic cues, promoting compact yet expressive representations, and 2) *cross-source adaptive fusion module* (CAFM) that performs cross-source attention weighting and local–global contextual refinement to enhance cross-source feature interaction. Experiments on three public benchmark datasets demonstrate that our RSCNet achieves superior performance compared with state-of-the-art methods, while maintaining substantially lower computational complexity. Our codes are publicly available at https://github.com/oucailab/RSCNet**

***Index Terms*—Hyperspectral image (HSI), multisource data fusion, sparse attention, spectral band, synthetic aperture radar (SAR).**

I. Introduction

RECENT years have witnessed the rapid development of satellite remote sensing sensors, leading to an increment in the availability of multisource remote sensing data. These multisource data provide strong support for various fields, such as land cover classification [1] [2], environmental monitoring [3], disaster prevention [4], natural resource management [5], hydrological monitoring [6], [7], and urban planning [8], [9]. Among these applications, land cover classification is a fundamental and important task [10], [11].

Hyperspectral images (HSI) play a key role in land cover classification, providing rich spectral information for different ground objects. HSI offers continuous spectral resolution, enabling detection of subtle features and identification of intrinsic properties for accurate land cover classification [12], [13], [14]. To further exploit this powerful spectral resolution, recent advances in hyperspectral analysis have delved into fine-grained feature representations [15], [16]. However, HSI data are sensitive to illumination, atmospheric effects, and cloud cover, leading to challenges in complex scenes. To address these issues, synthetic aperture radar (SAR) and light detection and ranging (LiDAR) data are often used as complementary data. SAR provides elevation, backscatter characteristics, and polarization signatures, along with all-weather imaging capabilities, enabling penetration of clouds and fog [17], [18], [19]. LiDAR data can provide a detailed 3-D structure of the terrain at high spatial resolution. However, SAR/LiDAR data lacks sufficient spectral information, limiting its ability to distinguish objects with similar appearances. The combination of HSI and SAR/LiDAR data enhances reliable land cover classification in diverse weather or seasonal scenarios. Therefore, in this study, we mainly focus on HSI and SAR/LiDAR data joint classification.

Due to differences in imaging principles between HSI and SAR/LiDAR data, effectively fusing these two heterogeneous data types presents a critical challenge [20]. Currently, many CNN and transformer-based methods have been proposed for multisource data classification [21], [22]. CNN-based methods [23], [24], [25] typically use two specific CNN encoders to extract spectral–spatial features from HSI and SAR/LiDAR data. Their feature maps are then aligned and fused via concatenation or attention to produce a unified representation for classification. Transformer-based methods [26], [27], [28] leverage separate spectral–spatial and geometric tokenizers to encode each data, and then apply cross-attention to exchange complementary information such as material signatures and elevation structure. The fused tokens are jointly modeled through global self-attention, enabling robust land-cover classification even under spectral redundancy and spatial misregistration.

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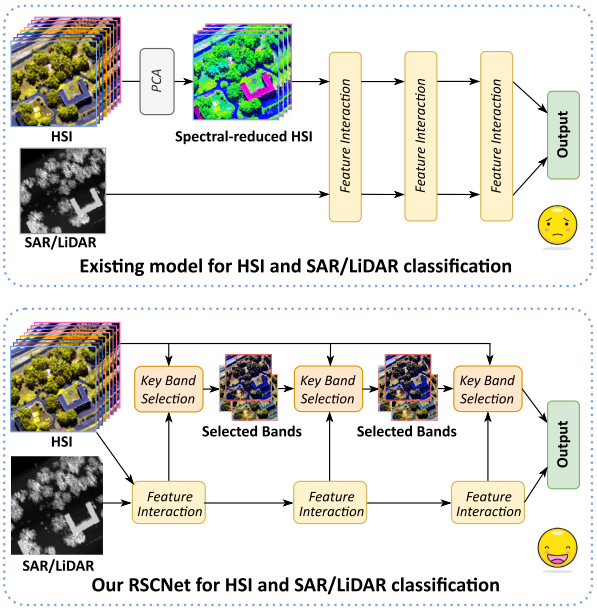


Fig. 1. Comparison between existing multisource data classification models and the proposed RSCNet. Existing methods apply task-agnostic PCA-based spectral reduction to HSI and then perform feature interaction with SAR/LiDAR, which may discard discriminative spectral information. In contrast, our RSCNet integrates progressive key band selection with cross-source feature interaction, allowing spectral bands to be dynamically refined under multisource guidance, thereby better preserving discriminative information and improving classification performance.

Despite their promising performance in land cover classification, existing deep learning-based methods still face the following two critical challenges.

1) *How to Reduce the spectral Redundancy Without Losing Multisource Complementary Features:* Hyperspectral sensor captures data across hundreds of contiguous spectral bands, and the rich spectral resolution introduces spectral redundancy. The spectral redundancy would increase computational cost and reduce the classification accuracy if noisy bands are not handled properly. Some methods use principal component analysis (PCA) for spectral dimensionality reduction [29], as shown in Fig. 1. However, most existing multisource fusion frameworks treat spectral reduction and multisource interaction as two isolated stages. This premature, task-agnostic dimensionality reduction prior to network training inevitably discards critical class-specific spectral signatures that might be highly complementary to SAR/LiDAR structural priors. In contrast to simple linear dimensionality reduction, advanced deep learning architectures, such as two-stream convolutional networks, have proven highly effective in automatically extracting abundant and task-specific spectral features [30]. Therefore, how to design a dynamic spectral selection mechanism that is jointly optimized with multisource fusion remains a significant hurdle.
2) *How to Enhance the Multisource Feature Interactions Under Severe Physical Heterogeneity:* HSI and SAR/LiDAR data are inherently heterogeneous with completely distinct imaging mechanisms. Traditional fusion methods (e.g., feature concatenation, feature summation, and cross attention) may fail to bridge the severe semantic gap and capture the nonlinear relationships between spectral and structural features, leading to suboptimal classification performance. Addressing nonlinear mappings is a pervasive issue in hyperspectral processing, where techniques combined with spatial–spectral adaptive strategies have shown great potential in constructing discriminative nonlinear feature spaces [31]. Inspired by these representation learning strategies, therefore, how to devise an adaptive strategy to align the semantic space between HSI and SAR/LiDAR data is a tricky task.

To address the above two challenges, we propose a representative spectral correlation network (RSCNet) for multisource remote sensing data joint classification. The core technical advantage of RSCNet lies in tightly coupling the dynamic spectral band selection with multisource feature interaction. As shown in Fig. 2, we design RSCNet with two key modules: *key band selection module* (KBSM) and *cross-source adaptive fusion module* (CAFM). To flexibly and dynamically exploit important spectral information, the KBSM filters redundant informative spectral bands from the HSI. Unlike traditional processing methods, KBSM explicitly leverages the structural and spatial priors from SAR/LiDAR as guidance to adaptively evaluate the importance of each spectral band. This mechanism ensures that the selected representative spectral band features are more valuable for the joint classification task. To enhance the multisource feature interactions and alleviate the heterogeneous semantic gap, we propose a CAFM. It first learns multisource attention to adaptively reweight and integrate heterogeneous features, enabling effective multisource interaction. It then combines local and global attention mechanisms to produce a discriminative fused representation that captures fine-grained spatial boundaries and long-range contextual dependencies.

Our main contributions can be summarized as follows.

1) We address the limitation of treating spectral redundancy reduction as an isolated processing step. We design an innovative KBSM that establishes a tight coupling between spectral reduction and multisource interaction. Unlike task-agnostic PCA-based reduction, KBSM adaptively selects the most informative spectral bands under the structural and spatial guidance of fused features from SAR/LiDAR. This mechanism effectively preserves critical class-specific spectral signatures while suppressing spectral redundancy.
2) We proposed the CAFM, which bridges the severe semantic gap caused by physical imaging heterogeneity. By combining the cross-source attention weighting mechanism with local–global contextual refinement, the CAFM dynamically reweights heterogeneous features and simultaneously enhances fine-grained spatial details and long-range contextual dependencies, yielding a more discriminative and consistent multisource representation.
3) Extensive experiments conducted on three benchmark datasets demonstrate that our proposed RSCNet significantly outperforms other state-of-the-art methods.

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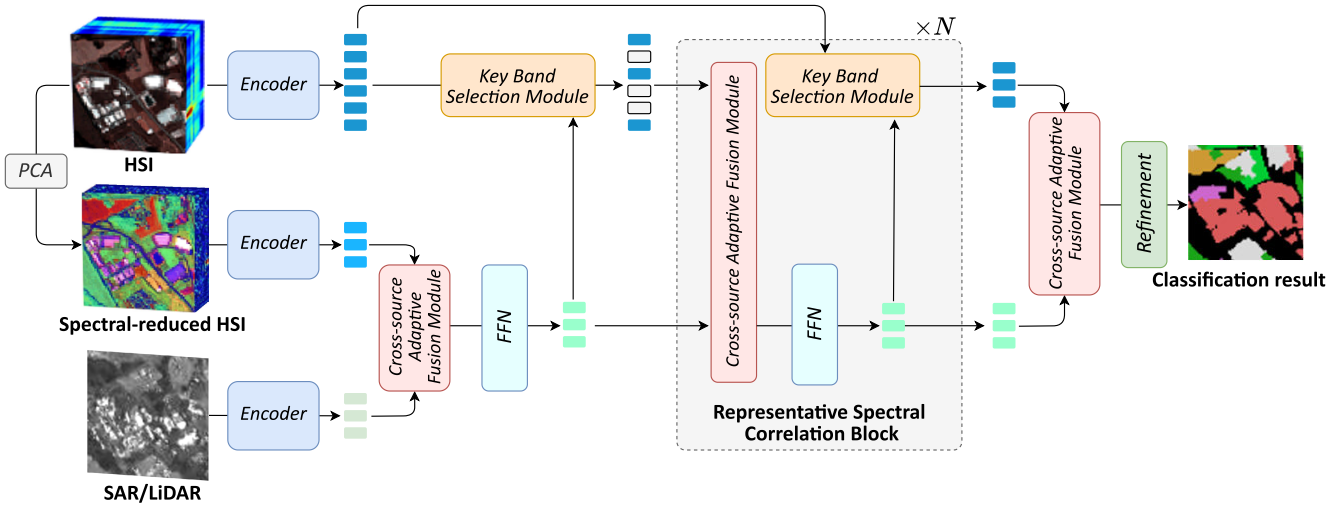


Fig. 2. Framework of the proposed RSCNet. The framework takes HSI, its PCA-reduced counterpart, and SAR/LiDAR data as inputs, which are individually encoded to obtain specific feature representations. The spectral-reduced HSI and SAR/LiDAR features are first integrated through the CAFM. Guided by the fused features, the KBSM dynamically selects the most informative spectral tokens. These selected tokens, together with the fused representation, are further processed within the RSCB. Finally, the refined multisource features are fused to produce the classification results.

We released the codes and parameters to facilitate the research community.

II. Related Works

A. Deep Learning-Based Multisource Remote Sensing Classification

Convolutional neural networks (CNNs) are widely used for feature extraction and fusion of multisource data. Yang et al. [23] proposed a single-stream CNN architecture employing separable dynamic grouping convolution (SepDG-Conv), enabling the network to learn and adjust its structure during training, more effectively integrating multisource data while maintaining high network efficiency. Gao et al. [24] developed a CNN based on feature exchange (FE-CNN) that initially extracts features from multisource data using multiple residual blocks, then implements feature exchange through batch normalization (BN) layers, thereby enhancing the accuracy of multisource data classification. Geng et al. [25] proposed a multisource joint representation learning method called multisource information bottleneck fusion network (MIBF-Net), which effectively integrates multisource information through mutual information (MI) constraints to generate comprehensive and nonredundant multisource representations. Song et al. [32] proposed a multiscale pseudo-Siamese network, incorporating attention mechanisms for the classification of hyperspectral and LiDAR data. Li et al. [33] introduced a new dual-channel spatial, spectral, and multiscale attention convolutional long short-term memory neural network (dual-channel A^3CLNN), designed with a special multiscale attention mechanism to learn enhanced spectral and spatial feature representations for feature extraction and classification of multisource remote sensing data. Zhao et al. [2] proposed the dynamic frequency feature fusion network (DFFNet), which utilizes a dynamic filter block to adaptively learn filter kernels in the frequency domain and incorporates a spectral–spatial adaptive fusion block to enhance the interaction and fusion of multisource data. To enhance open-set recognition of hyperspectral and LiDAR data, Xi et al. [34] proposed HyLiOSR, an autoencoder framework using a staged progressive learning strategy that integrates multiple Gaussian prototypes for known classes and a virtual unknown classifier for synthesized unknowns.

Despite their effectiveness, CNN-based models are inherently limited by local receptive fields, making it difficult to capture long-range dependencies and global contextual relationships across heterogeneous sources. To address these limitations, transformer-based methods have recently been introduced into multisource image classification. Xu et al. [35] proposed a joint convolutional cross ViT Network, which flexibly constructs local and global nonlinear feature mappings through the integration of multiscale convolution and transformer. Wang et al. [26] proposed a multimodal transformer cascaded fusion net (MMTCFN), which fully utilizes deep feature fusion through cascaded CNN and transformer modules. Li et al. [36] proposed the multifeature cross attention-induced transformer network (MCAITN), which enhances the classification accuracy of HSI and LiDAR data through a cross-attention module. Ma et al. [27] proposed the intra- and intersource interactive representation learning network (I^3RL-Net), which addresses the issue of information redundancy among multisource data by enhancing the dependencies within multisource features and performing enhanced fusion between sources. Yao et al. [28] proposed the extended vision transformer (ExViT), which enhances feature fusion through the cross-modality attention (CMA) module. Song et al. [37] proposed the binary-tree transformer network (BTRF-Net), achieving multimodal feature fusion through a binary tree structure and enhancing the network's stability and robustness. Lin et al. [38] proposed the dynamic cross-modal feature interaction network (DCMNet), which introduces a dynamic routing mechanism to construct data-dependent computational

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paths, adaptively capturing complementary features between HSI and LiDAR data. Jiao et al. proposed a dynamic common and unique feature fusion network (DCU-Net) [39], employing a multiscale attribute feature extraction block to capture initial spectral–spatial and elevation information. It utilizes a common-unique transformer (CUT) with cross-dynamic-agent-attention (cross-DAA) to explicitly extract shared and complementary information, which are then adaptively reconstructed into discriminative representations via a common and unique feature fusion (CUFF) block.

While the aforementioned methods have significantly advanced the state-of-the-art in multisource feature interaction, they predominantly treat hyperspectral dimensionality reduction as a static, isolated preprocessing step prior to fusion, which inevitably risks discarding critical class-specific signatures. To overcome this limitation, our RSCNet breaks the isolation between spectral reduction and multisource interaction. By employing the KBSM, it dynamically selects representative spectral bands under cross-source guidance, thereby explicitly preserving discriminative spectral information for optimal fusion.

B. Top-k Driven Sparse Attention Mechanism

Through the Top-k driven sparse attention mechanism, it is possible to effectively extract the most critical content and enhance classification accuracy. In the computer vision community, Chen et al. [40] proposed an effective deraining sparse transformer (DRSformer), which can adaptively retain the most valuable attention weights for feature aggregation. Jiang et al. [41] proposed the multiscale Top-k rank preservation, which achieves robust feature matching for images. Li et al. [42] proposed FlexAttention, a mechanism designed for efficient high-resolution vision–language models. It dynamically selects the most relevant regions in an image through Top-k retrieval based on the computed attention values between image and text tokens. This enables the model to focus on the critical areas of the image, thereby achieving more accurate results. Xiao et al. [43] proposed the Top-k token selective transformer (TTST), which dynamically selects the highest scoring Top-k keys, selectively focusing on the most critical tokens, thereby significantly enhancing feature aggregation capabilities and model performance. Shang et al. [44] proposed semantic sparse attention (SSA) to enhance texture details. SSA employs a Top-k selection strategy constrained by semantic masks, forcing the model to attend exclusively to the most matching style feature points. This sparse mechanism effectively avoids the smoothing effect of global weighting. By leveraging the Top-k retrieval mechanism, these approaches effectively resolve the distraction caused by irrelevant global information. This selective attention mechanism ensures a more precise aggregation of critical features, fundamentally strengthening the model's ability to handle complex scenarios with higher accuracy and robustness.

Our RSCNet introduces the Top-k driven attention mechanism to multisource remote sensing data classification. With this mechanism, our RSCNet automatically selects the most discriminative features, effectively reducing noise while retaining crucial information, thereby enhancing classification performance.

III. Methodology

In this section, we provide the implementation details of the proposed RSCNet for multisource data joint classification. After that, we will provide an in-depth explanation of the KBSM and CAFM.

A. Overall Framework of the RSCNet

As shown in Fig. 2, we apply PCA for spectral dimensionality reduction on the original HSI, resulting in a spectral-reduced HSI. The original HSI, spectral-reduced HSI, and SAR/LiDAR data are then fed into the encoders for feature extraction. This process yields the HSI features $\mathbf{F}_h \in \mathbb{R}^{hw\times c}$, the spectral-reduced HSI features $\mathbf{F}_r \in \mathbb{R}^{hw\times r}$, and the SAR/LiDAR features $\mathbf{F}_x \in \mathbb{R}^{hw\times r}$. Here, c represents the number of spectral bands in the original HSI, r denotes the number of spectral bands in the spectral-reduced HSI, and hw indicates the feature dimension of each band.

The spectral-reduced HSI features $\mathbf{F}_r$ and SAR/LiDAR features $\mathbf{F}_x$ are fused via the CAFM, yielding the fused feature $\mathbf{F}_{fus}$. $\mathbf{F}_{fus}$ is fed into a feed-forward network (FFN) for nonlinear transformation. Next, the KBSM is employed to select the most important spectral tokens $\mathbf{F}'_h$ from $\mathbf{F}_h$. It is achieved by computing sparse attention between $\mathbf{F}_h$ and $\mathbf{F}_{fus}$.

Afterward, the selected spectral tokens $\mathbf{F}'_h$, together with $\mathbf{F}_h$ and $\mathbf{F}_{fus}$, are fed into the representative spectral correlation block (RSCB) for further cross-source feature fusion to yield $\mathbf{F}'_{fus}$. The RSCB serves as the core component for iterative cross-source feature interaction and refinement. Within each RSCB, the fused multisource representation from the CAFM is first enhanced through FFN to capture nonlinear spectral–spatial correlations. Meanwhile, the KBSM dynamically identifies the most informative spectral tokens from the original HSI. These selected tokens are then reintegrated with the fused features to strengthen cross-source alignment and mitigate redundancy. The RSCB is repeated N times for cross-source feature optimization, and N is an important parameter that will be analyzed in the experiments. Finally, the output of RSCB is fused via cross-attention, and a two-layer MLP is utilized to generate the final classification results.

B. Key Band Selection Module

The details of the KBSM are shown in Fig. 3. First, the hyperspectral features of size $hw \times c$ are fused with the fused feature representation $hw \times k$ via matrix multiplication, producing an attention map of size $c \times k$. A linear transformation is then applied to aggregate information across channels, resulting in a compact vector of size $c \times 1$. Next, a dynamic sparse gating module (DSGM) is used to adaptively determine the number of informative spectral bands. It generates a sparse mask vector by selecting only the most relevant k spectral bands in a data-dependent manner. The selected band indices are then used to filter the original hyperspectral features, yielding an output of size $hw \times k$ that contains only the retained selected spectral bands. This design efficiently focuses

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Algorithm 1 Key Band Selection Module

Require: Hyperspectral features $X \in \mathbb{R}^{hw\times c}$, auxiliary fused features $Z \in \mathbb{R}^{hw\times k}$

Ensure: Selected spectral-band features $Y \in \mathbb{R}^{hw\times k}$

Attention Map Construction:

$A = X^{\top}Z \in \mathbb{R}^{c\times k}$ ▷Matrix multiplication

Attention Vector Generation:

$v = \text{Linear}(A) \in \mathbb{R}^{c\times 1}$

Dynamic Sparse Gating Module:

$g = \sigma(\text{MLP}(X)) \in \mathbb{R}^{c\times 1}$ ▷Sigmoid activation

$\hat{v} = v \odot g$ ▷Element-wise spectral gating

$s = \text{Top-}k(\hat{v})$ ▷Select indices of informative bands

Band Selection:

$Y = \text{Select}(X, s) \in \mathbb{R}^{hw\times k}$

return: Y

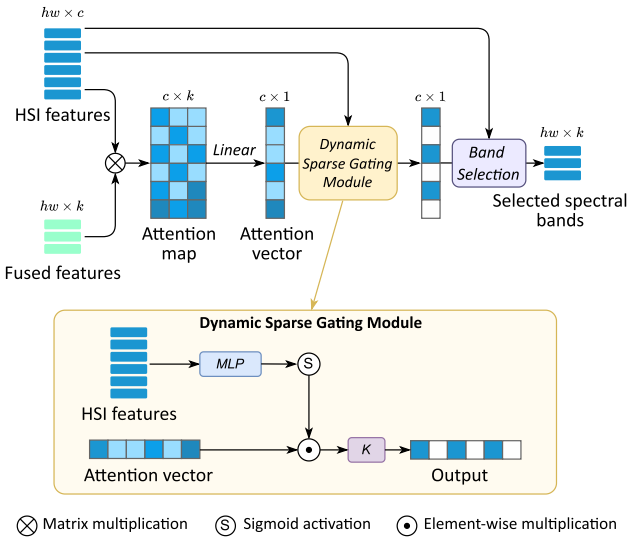


Fig. 3. Details of KBSM. HSI features and fused features are combined via matrix multiplication, and then the fused information is compressed into a compact vector. A DSGM adaptively selects the most informative spectral bands, retaining only key channels to reduce redundancy and enhance spectral–spatial feature learning.

computation on the most informative spectral bands while reducing redundancy, helping the network better exploit the spectral–spatial correlations in multisource data. Details of the KBSM are shown in Algorithm 1.

Details of the DSGM are illustrated in Fig. 3. The HSI features are first fed into an MLP, followed by a sigmoid activation function to produce a gating vector. This gating vector is then multiplied by the attention vector via element-wise multiplication to generate weighted responses. A sparsity operator selects the most informative k values from the weighted responses.

From a physical perspective, although HSI and SAR/LiDAR possess distinct imaging mechanisms, they exhibit strong complementary correlations when representing complex land covers [45]. The physical plausibility of utilizing SAR/LiDAR features to guide hyperspectral band selection lies in providing

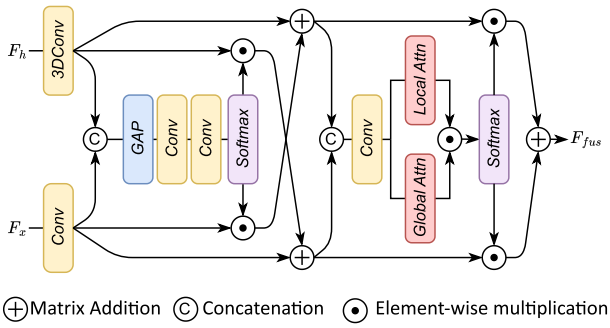


Fig. 4. Illustration of the CAFM. It first learns cross-source attention to adaptively weight HSI and SAR/LiDAR features, and then applies local–global attention to refine spatial details and contextual information, yielding robust fused feature representation.

structural and spatial priors for spectral feature extraction. Specifically, different physical materials may share similar spectral signatures, causing certain hyperspectral bands to become redundant or introduce confusion. However, these materials may exhibit completely distinct elevation and backscattering mechanisms. By injecting SAR/LiDAR features as guidance, the KBSM leverages structural context to inform the network of the target's geometric nature, thereby dynamically assigning higher weights to the spectral bands that are most discriminative for that specific structural category. Furthermore, in areas where HSI becomes physically unreliable due to shadows or illumination variations, the illumination-independent SAR/LiDAR features can guide the network to suppress noisy spectral bands, thus ensuring that the band selection process is both physically interpretable and robust.

The design of KBSM is rooted in the essential distinction between redundant correlation and discriminative correlation structures within hyperspectral data. Typically, adjacent continuous bands exhibit high linear statistical dependency (i.e., redundant correlation), sharing duplicated low-level information that limits feature diversity. Conversely, specific task-relevant bands discretely distributed across the spectrum can form high-order, nonlinear complementary relationships (i.e., discriminative correlation structures). Therefore, rather than simply discarding features, the KBSM is specifically designed to break the statistical redundancy while actively mining and aggregating these discriminative correlation structures, thereby yielding a compact yet highly expressive representation.

C. Cross-Source Adaptive Fusion Module

To effectively integrate heterogeneous information from HSI and SAR/LiDAR data, we propose a CAFM, as illustrated in Fig. 4. The module exploits complementary spectral–spatial representations from HSI and structural cues from SAR/LiDAR data through two cascaded stages: cross-source attention weighting and local–global contextual refinement.

1) Cross-Source Attention Weighting: Given the HSI feature F_h and the SAR/LiDAR feature F_x, we first transform them using a 3-D convolution and a 2-D convolution, respectively, to project the two sources into a unified embedding

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space while preserving their inherent characteristics. The transformed features are then concatenated and passed through a global average pooling (GAP) operator and two convolution layers to derive attention descriptors. A Softmax activation generates normalized attention weights that adaptively measure the contribution of each source. These weights are multiplied back to the individual source features, enabling cross-source enhancement where more informative source cues are emphasized. The weighted features are summed to derive an intermediate fused representation.

2) Local–Global Contextual Refinement: Although the above step enables source-level balancing, the fused features still lack fine contextual interactions. To address this issue, we employ a local–global dual attention unit. Specifically, a convolution layer first enriches the fused features with contextual encoding. Then, we simultaneously deploy local and global attention. Local attention focuses on detailed spatial texture patterns beneficial for edge and object boundary preservation. Global attention models long-range dependencies and suppresses local noise and redundancy. The outputs of the two branches are multiplied, followed by a Softmax activation to ensure stable attention distribution. The resulting product is then multiplied by the intermediate fused features for content-adaptive refinement. Finally, the two features are summed to yield the final fused feature F_{fus}.

The CAFM provides a flexible mechanism for robust fusion of HSI and SAR/LiDAR data, leading to more discriminative and consistent representations for spectral band selection and cross-source classification.

IV. Experimental Results and Analysis

A. Experimental Setup and Dataset Description

To validate the effectiveness of our proposed RSCNet, we conducted experiments on three multisource remote sensing datasets: Augsburg [46], Berlin [47], and Houston2013 [48]. The Augsburg and Berlin datasets are used for the classification of HSI and SAR data. The Houston2013 datasets are used for the classification of HSI and LiDAR data. All models are implemented in the PyTorch framework and trained using an NVIDIA RTX 4090 GPU with an Intel Xeon Platinum 8474C processor.

The Augsburg dataset consists of HSI and SAR data from Augsburg, Germany. The HSI data were captured by the HySpex sensor, containing 180 spectral bands covering a spectral range from 0.4 to 2.5 μm. The SAR data were captured by the Sentinel-1 sensor, and the SAR data have four features derived from polarization decomposition (VV intensity, VH intensity, real part, and imaginary part of the off-diagonal element of the PolSAR covariance matrix). All images have a ground sample distance (GSD) of 30 m. The spatial size of both images is 332×485 pixels.

The Berlin dataset covers urban and rural areas of Berlin. The HSI data are simulated using HyMap HS data to mimic the environmental mapping and analysis program (EnMAP) data. The SAR data are provided by the European Space Agency (ESA) as Sentinel-1 dualPol (VV–VH) single look complex (SLC) products. The HSI image has 244 spectral bands, covering a spectral range from 0.4 to 2.5 μm [47], with a spatial size of 797×220 pixels. The SAR image has a spatial size of 1723×476 pixels. Nearest-neighbor interpolation was used to match the spatial resolution of the two images.

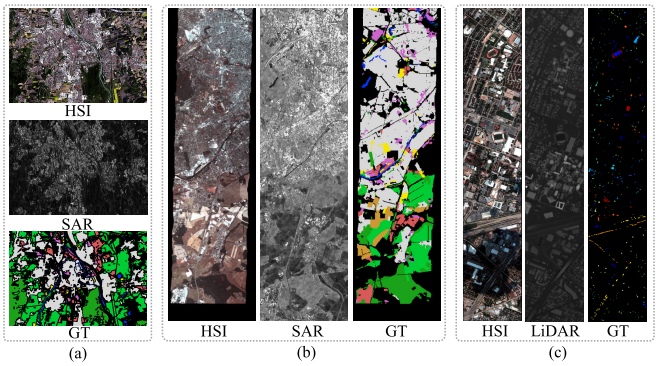


Fig. 5. Visualization of the datasets. (a) Augsburg dataset. (b) Berlin dataset. (c) Houston2013 dataset.

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THE AUGSBURG DATASET

No.	Name	Color	Train	Test
1	Forest		146	13361
2	Residential area		264	30065
3	Industrial area		21	3830
4	Low plants		248	26609
5	Allotment		52	523
6	Commercial area		7	1638
7	Water		23	1507
	Total		761	77533

The Houston2013 dataset originates from the 2013 IEEE Geoscience and Remote Sensing Society Data Fusion Contest. It provides urban land cover for the city of Houston, Texas, and its surrounding areas. The spatial resolution of the dataset is 2.5 m, and its spatial size is 349×1905 pixels. The hyperspectral imagery encompasses 144 spectral bands covering wavelengths from 380 to 1050 nm. The average altitude of the sensor above the ground was 5500 ft. The dataset also includes LiDAR data. The average altitude of the sensor above the ground was 2000 ft. It encompasses 15 different types of land cover. The dataset was acquired by the National Science Foundation (NSF)-funded National Center for Airborne Laser Mapping (NCALM) at the University of Houston campus and its surrounding areas.

Tables I–III list the number of training and testing samples for the three datasets. Fig. 5 shows the false-color composite HSI data, SAR/LiDAR images, and ground truth images. The evaluation metrics used in this article include overall accuracy (OA), average accuracy (AA), and Kappa. Specifically, OA represents the proportion of correctly classified samples in the entire dataset, AA represents the average accuracy of all classes present in the dataset, and Kappa is used as a statistical measure of agreement between predicted labels and true labels.

B. Analysis of Important Parameters

1) Key Band Ratio in KBSM: The key band ratio, denoted as K, is an important parameter in the KBSM that directly

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TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR THE BERLIN DATASET

No.	Name	Color	Train	Test
1	Forest		443	54511
2	Residential area		423	268219
3	Industrial area		499	19067
4	Low plants		376	58906
5	Soil		331	17095
6	Allotment		280	13025
7	Commercial area		298	24526
8	Water		170	6502
	Total		2820	461851

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR THE HOUSTON2013 DATASET

No.	Name	Color	Train	Test
1	Health grass		198	1053
2	Stressed grass		190	1064
3	Synthetic grass		192	505
4	Trees		188	1056
5	Soil		186	1056
6	Water		182	143
7	Residential		196	1072
8	Commercial		191	1053
9	Road		193	1059
10	Highway		191	1036
11	Railway		181	1054
12	Parking lot 1		192	1041
13	Parking lot 2		184	285
14	Tennis court		181	247
15	Running track		187	473
	Total		2832	12197

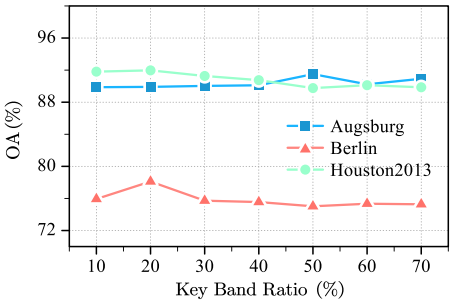


Fig. 6. Effect of the key band ratio on classification performance on three datasets.

determines the feature dimensionality k of the selected spectral bands. The dimension k is dynamically scaled by the ratio K, which consequently affects the classification performance. We select different Key Band Ratios ranging from 10% to 70%, and the experimental results are shown in Fig. 6. As shown in the figure, the OA values on Augsburg and Houston2013 remains consistently high. Our RSCNet achieved optimal performance on the Augsburg dataset when $K = 50\%$, and on the Houston2013 dataset when $K = 20\%$. In contrast, the Berlin dataset shows slightly more noticeable variation, yet still maintains a relatively stable trend. Our RSCNet achieves the best performance when K is set to 20% on the Berlin dataset. Therefore, in our following experiments, K is set to

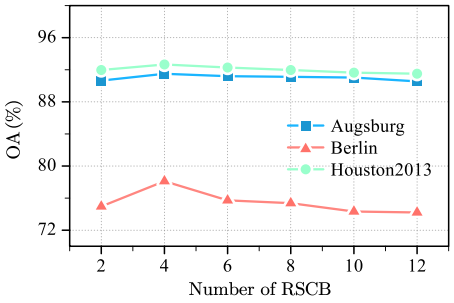


Fig. 7. Effect of the number of RSCB on classification performance on three datasets.

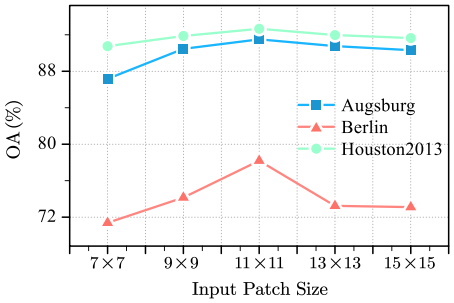


Fig. 8. Effect of the input patch size on classification performance on three datasets.

50% on the Augsburg dataset, while on the other datasets, K is set to 20%.

2) Number of RSCB: The RSCB iteratively reinforces cross-source complementary and discriminative spectral cues throughout the network. We tested various number of RSCB N, ranging from 2 to 12. The experimental results are shown in Fig. 7. As can be seen, increasing the number of RSCBs generally leads to stable or slightly improved performance, where the optimal results are typically achieved when we use four blocks. Beyond this range, the accuracy tends to plateau or even slightly decline, suggesting that excessively stacking RSCBs may introduce redundant feature interactions and increase the computational burden. These results demonstrate that when N is set to 4, RSCBs are sufficient to capture representative spectral correlations and ensure strong classification performance.

3) Input Patch Size: As shown in Fig. 8, we tested five different input patch sizes, ranging from 7 to 15. As can be seen, increasing the patch size from 7 to 11 consistently improves the OA, suggesting that a moderate spatial context is beneficial for feature extraction. However, further enlarging the patch size to 13 and 15 brings no significant gain and even results in slight performance degradation. It is likely due to the introduction of redundant background information. Overall, the results indicate that 11×11 provides the most balanced spatial context and yields the best performance.

C. Performance Comparison

We compared our proposed RSCNet with ten state-of-the-art methods, including FusAtNet [49], S^2ENet [50], AsyFFNet [51], ExViT [28], HCT [52], MACN [53], MICF-Net [54], GCCQTNet [55], CHNet [56], and MGMNet [57]. FusAtNet

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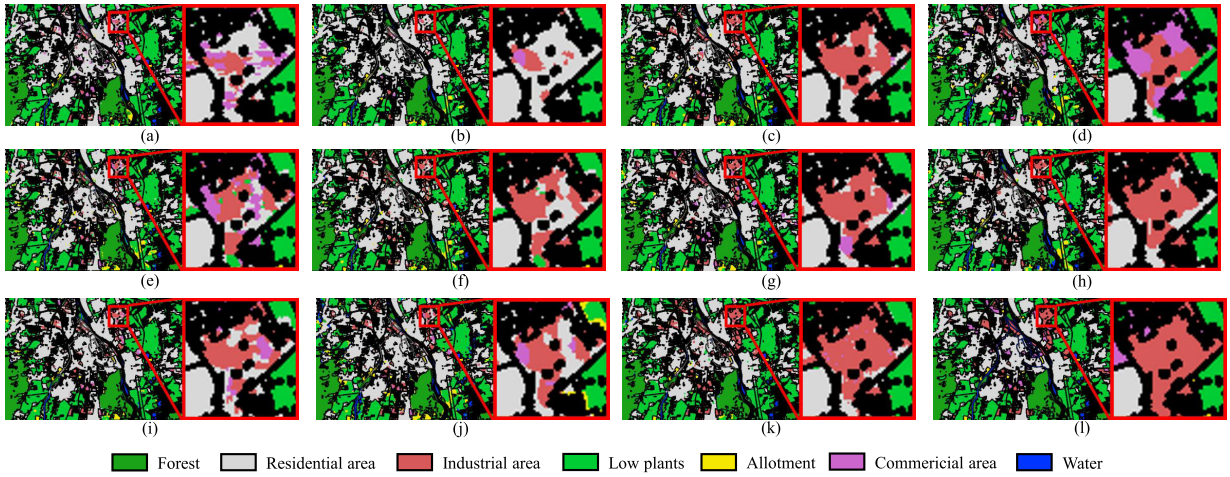


Fig. 9. Visualized classification results of different methods for the Augsburg dataset. (a) FusAtNet. (b) S^{2}ENet. (c) AsyFFNet. (d) ExViT. (e) HCT. (f) MACN. (g) MICF-Net. (h) GCCQTNet. (i) CHNet. (j) MGMNet. (k) Proposed RSCNet. (l) Ground truth.

TABLE IV

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE AUGSBURG DATASET. THE BOLD AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS

Class	FusAtNet	S^2ENet	AsyFFNet	ExViT	HCT	MACN	MICF-Net	GCCQTNet	CHNet	MGMNet	RSCNet
Forest	93.78	**98.10**	96.01	90.04	96.92	97.95	95.98	95.26	96.24	96.28	96.07
Residential area	97.58	99.08	97.96	95.44	98.96	98.68	98.46	96.80	**99.22**	97.76	96.40
Industrial area	26.48	12.19	53.29	34.58	33.05	45.30	58.38	68.20	44.23	49.61	**78.38**
Low plants	**97.67**	91.78	91.74	90.68	92.36	94.24	91.75	91.52	93.83	92.32	94.08
Allotment	52.77	45.12	**65.39**	51.82	54.11	51.24	40.73	41.68	54.68	56.60	58.32
Commercial area	24.66	1.22	4.58	**28.63**	6.62	6.74	23.53	13.97	19.85	5.15	1.96
Water	47.51	24.09	49.64	17.65	4.74	49.27	44.81	46.80	46.74	49.93	**50.07**
OA(%)	90.62	88.22	90.15	86.65	89.83	91.18	90.74	90.22	91.14	90.10	**91.50**
AA(%)	62.92	53.08	65.51	58.40	61.34	63.35	64.81	64.89	64.97	63.95	**67.89**
Kappa	0.8633	0.8647	0.8574	0.8079	0.8528	0.8723	0.8665	0.8600	0.8710	0.8575	**0.8776**

[49] employs a self-attention mechanism to process spectral features of HSI, while utilizing attention maps derived from LiDAR data to enhance spatial features. S^{2}ENet [50] introduces the spatial–spectral cross-source enhancement module, which enables cross-source information interaction between hyperspectral and LiDAR data. AsyFFNet [51] adopts an asymmetric feature fusion strategy that focuses on managing the asymmetry of information contributions from multisource data. ExViT [28] extends the traditional vision transformer (ViT) to handle multisource data using separable convolutions and integrates these features through a cross-attention module to enhance feature interaction. HCT [52] employs a dual-branch architecture, combining hierarchical CNNs with transformers. It uses a cross-token attention fusion encoder to jointly classify hyperspectral and LiDAR data. MACN [53] proposes the mixing of a self-attention and a convolution transformer layer, which achieves perception of both local and global information. MICF-Net [54] presents a dual-branch cross-source attention fusion transformer to extract global context and mine spatial relationship similarities between HSI and LiDAR data. It also introduces an adaptive mask modulation module to balance learning rates across sources. GCCQTNet [55] proposes an independent squeeze-expansion-like fusion (ISEF) structure to reduce multimodal heterogeneity using global information as an agent. Additionally, it introduces a cross-memory quaternion transformer (CMQT) to extract discriminative fusion features by modeling complex intra- and intermodality relationships. CHNet [56] proposes a spectral redundancy removal (SRR) module and a spatial reconstruction (SR) module to filter out multimodal redundancy in the frequency and spatial domains. Furthermore, it introduces a dynamic feature selection and fusion (DFSF) module and a factorized high-order feature learning (FHFL) module to adaptively integrate features and enhance their discriminability while reducing dimensionality. MGMNet [57] proposes a mutual-guidance mechanism embedded into a two-stream network to explore information interaction, where image features from one source are convolved with context-aware filters generated from another source.

1) Results on the Augsburg Dataset: Classification results of different methods are listed in Table IV. The corresponding visualized classification maps are shown in Fig. 9. Our RSCNet consistently achieves the highest or second-highest accuracy in most categories, particularly excelling in classes such as Industrial area, Allotment, and Water. It also delivers the best overall performance, achieving 91.50% in OA, 67.89% in AA, and a Kappa of 0.8776, outperforming all competing baselines. These results demonstrate that RSCNet provides superior generalization and class discrimination capabilities compared to existing CNN- and transformer-based approaches.

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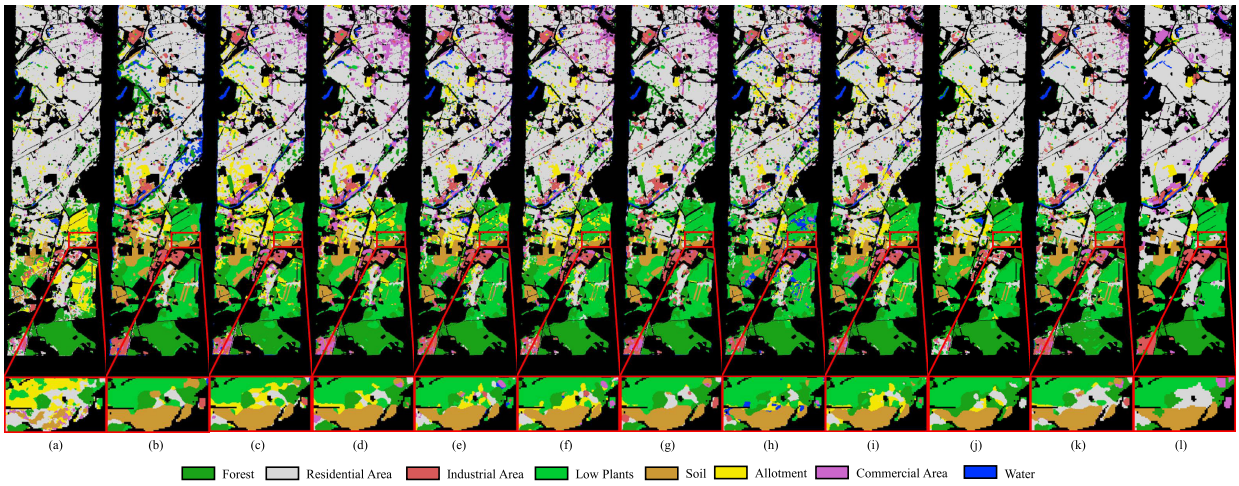


Fig. 10. Visualized classification results of different methods for the Berlin dataset. (a) FusAtNet. (b) S^2ENet. (c) AsyFFNet. (d) ExViT. (e) HCT. (f) MACN. (g) MICF-Net. (h) GCCQTNet. (i) CHNet. (j) MGMNet. (k) Proposed RSCNet. (l) Ground truth.

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE BERLIN DATASET. THE BOLD AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS

Class	FusAtNet	S^2ENet	AsyFFNet	ExViT	HCT	MACN	MICF-Net	GCCQTNet	CHNet	MGMNet	RSCNet
Forest	86.24	81.09	77.40	78.01	81.89	79.23	81.77	87.46	84.84	**90.20**	73.52
Residential area	**91.38**	73.05	71.30	74.05	76.04	80.27	80.28	77.74	81.14	88.52	88.07
Industrial area	19.76	**62.61**	38.98	39.48	54.14	42.99	58.87	54.34	56.53	22.84	46.60
Low plants	20.00	82.82	79.30	84.15	85.83	79.95	**88.98**	71.62	78.24	74.46	83.42
Soil	48.72	86.41	87.69	88.03	77.70	76.98	84.13	72.80	79.32	75.42	**90.72**
Allotment	38.89	54.61	**81.65**	70.00	74.93	76.21	31.94	67.39	74.53	54.24	24.35
Commercial area	18.47	2.56	36.52	**38.18**	35.79	31.30	20.32	20.42	17.67	3.38	12.99
Water	29.61	75.96	75.96	56.41	63.46	57.55	52.41	57.94	63.80	9.64	**76.13**
OA(%)	70.91	71.08	70.82	72.63	74.79	75.41	75.88	73.34	76.32	77.13	**78.19**
AA(%)	44.13	64.88	68.59	66.04	**68.72**	65.56	62.34	63.71	67.01	52.34	61.98
Kappa	0.5107	0.5802	0.5853	0.6048	0.6303	0.6315	0.6350	0.6052	0.6444	0.6243	**0.6544**

In Fig. 9, we can see that RSCNet demonstrated superior classification performance in the Industrial area. The findings of this experiment significantly reveal the exceptional local feature extraction capabilities and long-range contextual modeling abilities of RSCNet when dealing with fine-grained categories. These results demonstrate that RSCNet provides superior generalization and class discrimination capabilities compared to existing CNN- and transformer-based approaches.

2) Results on the Berlin Dataset: Table V illustrates the classification performance of different methods on the Berlin dataset, and Fig. 10 shows the corresponding classification maps. RSCNet achieves the highest scores in Soil and Water, demonstrating its strong capability in handling complex and fine-grained classes. Although a few methods exhibit slightly higher results in certain categories, RSCNet maintains the most balanced and robust performance across the dataset, validating its effectiveness and generalization on multisource data classification.

3) Results on the Houston2013 Dataset: The classification performance of different methods on the Houston2013 dataset is shown in Table VI and Fig. 11. Overall, most models achieve strong results on this dataset, but our RSCNet demonstrates clear advantages in terms of both global metrics and per-class stability. RSCNet secures the best OA value of 92.66%, outperforming all competing methods. Notably, our method demonstrates significant improvements in challenging categories such as Trees, Highway, and Railway. RSCNet has demonstrated its significant advantages in spatial feature extraction and spectral feature extraction. By leveraging these features comprehensively, RSCNet is not only capable of identifying the basic contours of objects but also capturing more subtle details. These results demonstrate that our RSCNet delivers the most balanced and robust classification capability on the Houston2013 dataset, validating its effectiveness and competitive edge in real-world complex scene understanding.

D. Ablation Study

1) Effectiveness of Essential Components: To evaluate the effectiveness of the essential components of our proposed RSCNet, we conducted ablation studies to explore the effectiveness of the KBSM and CAFM. The experimental results are illustrated in Table VII. It is noteworthy that in experiments without KBSM, random selection was used as a substitute. In experiments where CAFM was not used, we used cross-attention instead. The baseline model achieves 90.19%, 72.27%, and 89.45% on three datasets,

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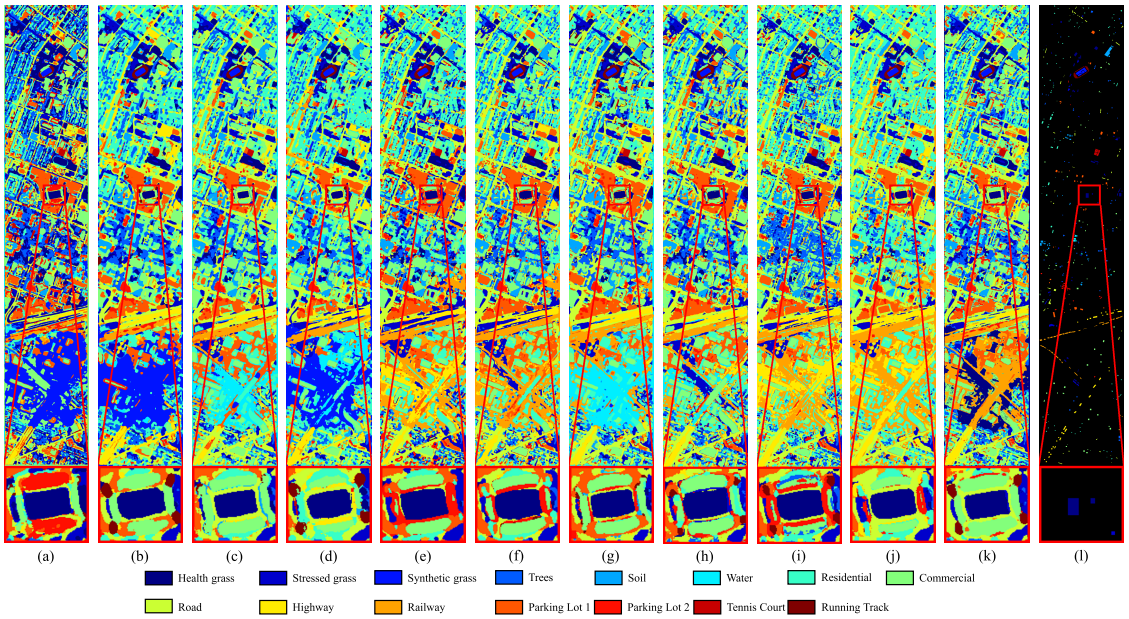


Fig. 11. Visualized classification results of different methods for the Houston2013 dataset. (a) FusAtNet. (b) S^2ENet. (c) AsyFFNet. (d) ExViT. (e) HCT. (f) MACN. (g) MICF-Net. (h) GCCQTNet. (i) CHNet. (j) MGMNet. (k) Proposed RSCNet. (l) Ground truth.

TABLE VI

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET. THE BOLD AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS

Class	FusAtNet	S^2ENet	AsyFFNet	ExViT	HCT	MACN	MICF-Net	GCCQTNet	CHNet	MGMNet	RSCNet
Health grass	82.53	81.48	82.72	82.72	81.10	82.90	82.91	82.05	**83.10**	83.00	82.53
Stressed grass	85.15	84.68	83.93	85.15	86.09	**100.0**	83.46	96.33	85.15	85.15	79.98
Synthetic grass	98.81	99.01	**100.0**	99.21	99.80	99.80	99.80	99.41	99.60	99.60	99.60
Trees	92.90	92.14	91.95	91.95	98.01	98.11	97.63	91.00	92.61	87.59	**99.15**
Soil	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	99.52	99.90	**100.0**	99.23
Water	98.60	**100.0**	95.80	98.60	95.10	95.80	95.80	95.80	95.80	89.51	98.60
Residential	85.45	89.93	**96.18**	86.66	83.49	78.17	83.49	90.95	88.25	95.62	86.57
Commercial	81.81	90.18	81.14	91.05	86.43	80.46	81.91	**95.67**	81.62	76.42	87.97
Road	83.76	92.63	86.12	91.12	86.69	90.93	82.15	87.54	89.61	**93.96**	90.93
Highway	53.67	64.86	64.58	64.86	79.05	55.88	66.22	65.15	73.84	85.14	**96.14**
Railway	79.65	97.60	86.76	77.06	95.97	**99.33**	95.59	86.85	97.41	94.91	**99.33**
Parking lot 1	91.26	88.09	90.59	88.76	98.46	99.71	94.62	99.14	**99.90**	98.37	96.93
Parking lot 2	81.05	91.93	**92.28**	87.72	90.88	88.77	81.40	82.81	89.47	89.12	89.47
Tennis court	**100.0**	**100.0**	95.14	97.17	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	99.60	**100.0**
Running track	98.73	**100.0**	**100.0**	98.52	**100.0**	**100.0**	99.79	**100.0**	**100.0**	**100.0**	**100.0**
OA	85.32	89.56	87.95	87.42	90.66	89.81	88.09	90.40	90.30	90.98	**92.66**
AA	87.56	91.50	89.81	89.37	92.07	91.33	89.65	91.48	91.75	91.87	**93.76**
Kappa	0.8412	0.8869	0.8696	0.8638	0.8986	0.8893	0.8712	0.8959	0.8947	0.9020	**0.9204**

TABLE VII

INFLUENCE OF KBSM AND CAFM ON CLASSIFICATION RESULTS OF RSCNET. THE BOLD DENOTE THE BEST RESULTS

KBSM	CAFM	Augsburg	Berlin	Houston2013
		90.19	72.27	89.45
	✓	91.13	73.35	91.31
✓		91.37	76.74	91.69
✓	✓	**91.50**	**78.19**	**92.66**

respectively. Introducing CAFM alone improves performance consistently on all datasets, while utilizing KBSM alone yields even larger gains, particularly on the Berlin dataset (from 72.27% to 76.74%). When both KBSM and CAFM are jointly incorporated, RSCNet obtains the highest accuracy on all datasets, demonstrating that both modules are complementary. These results confirm that integrating both components is crucial for maximizing the model's classification capability and overall robustness.

2) Feature Visualization: To qualitatively evaluate the effectiveness of the proposed KBSM, we employ t-SNE [58] to visualize the feature distributions on the Houston2013 dataset. Fig. 12(a) illustrates the visualization of the original HSI features, where samples from different categories are heavily overlapped and scattered, indicating significant spectral redundancy and lower discriminative power. In contrast, as shown in Fig. 12(b), the features filtered by our KBSM exhibit much clearer cluster boundaries, with significantly improved

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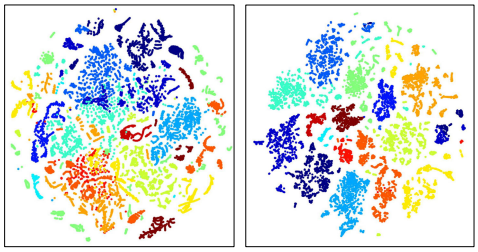


Fig. 12. Comparison of feature separability using t-SNE visualization. (a) Original HSI features with high redundancy. (b) HSI features after KBSM.

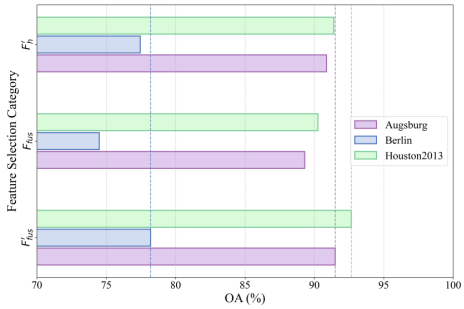


Fig. 13. Feature fusion validation on three datasets.

TABLE VIII

COMPARISON OF SPECTRAL REDUNDANCY AND DISCRIMINABILITY BEFORE AND AFTER KBSM PROCESSING. THE BOLD DENOTE THE BEST RESULTS

Features	Augsburg		Berlin		Houston2013	
	ACC ↓	MI ↑	ACC ↓	MI ↑	ACC ↓	MI ↑
Original HSI features	0.4342	0.4192	0.6537	0.3241	0.7452	0.7914
Selected band features by KBSM	**0.3732**	**0.4495**	**0.6172**	**0.3569**	**0.7179**	**0.8308**

TABLE IX

ANALYSIS ON THE EFFECT OF PCA PREPROCESSING ACROSS THREE DATASETS. THE BOLD DENOTE THE BEST RESULTS

Method	Augsburg		Berlin		Houston2013	
	OA (%)	Params (M)	OA (%)	Params (M)	OA (%)	Params (M)
w/o PCA	91.15	3.1243	76.32	5.7296	91.75	2.1204
PCA	**91.50**	**2.8812**	**78.19**	**5.2596**	**92.66**	**1.8527**

TABLE X

COMPARISON OF OA (%) AMONG DIFFERENT BAND SELECTION/DIMENSIONALITY REDUCTION METHODS ON THE BERLIN DATASET

Method	OA (%)
ISSC	74.50
OPBS	75.74
BAM	76.51
SRL-SOA	76.08
LGCAF	77.43
KBSM (Ours)	**78.19**

interclass separability and intraclass compactness. This comparison demonstrates that the KBSM effectively identifies the most informative bands and suppresses noisy or redundant spectral information, thereby enhancing the model's ability to learn more representative and discriminative features for classification.

3) Analysis of Spectral Redundancy and Discriminability: To analyze the impact of KBSM on spectral redundancy and feature discriminability, we evaluated the original HSI features and the selected band features using two complementary metrics: average correlation coefficient (ACC) and MI. Specifically, ACC calculates the average Pearson correlation coefficient among all available band features, providing a clear measure of low-level linear statistical redundancy. In contrast, MI computes the dependency between band features and the ground-truth labels, quantifying the nonlinear, task-relevant discriminative capacity. Experiments were conducted on all three datasets, and the results are presented in Table VIII. After processing with KBSM, the ACC of the selected band subsets shows a downward trend, confirming the successful suppression of linear redundant dependencies among adjacent bands. Simultaneously, the average MI scores exhibit an upward trend, indicating that the retained band subsets establish a stronger correlation with the semantic labels and contain richer discriminative information. These results demonstrate that KBSM effectively breaks useless statistical redundancy while actively aggregating specific bands with high discriminability.

4) Feature Fusion Validation: To validate the effectiveness of feature fusion, we use three feature sets: The first set, defined as the key spectral tokens $\mathbf{F}'_h$, comprises the top-k discriminative spectral bands extracted directly from the original HSI to preserve high-fidelity spectral signatures. The second set consists of the intermediate cross-source features $\mathbf{F}_{fus}$, which are derived from the integration of auxiliary SAR/LiDAR data and PCA-reduced HSI via the CAFM. Finally, the fused features $\mathbf{F}'_{fus}$ represent the ultimate refined multisource representation, generated by further integrating $\mathbf{F}'_h$ and $\mathbf{F}_{fus}$ through an additional pass in the CAFM. The classification results are shown in Fig. 13. As can be observed, the final fused features $\mathbf{F}'_{fus}$ consistently achieve the highest OA values across all three datasets, demonstrating the clear benefit of combining key spectral tokens and cross-source representations. The performance obtained using only the key spectral tokens $\mathbf{F}'_h$ ranks second, outperforming the intermediate cross-source features $\mathbf{F}_{fus}$. This is particularly evident on the Berlin dataset, where $\mathbf{F}_{fus}$ exhibits the lowest OA value due to information loss introduced by PCA-based dimensionality reduction. In contrast, $\mathbf{F}'_{fus}$ leverages both high-fidelity spectral signatures and structural cues from auxiliary sources, yielding a notable improvement. Overall, the experimental results confirm the complementary nature of both sources and validate the effectiveness of the proposed fusion strategy.

5) Effectiveness of PCA Preprocessing: To clarify the role of PCA in our architecture, we conduct an experiment by omitting the PCA preprocessing, where the original HSI is directly fed into the CAFM using a standard convolutional mapping instead. As shown in Table IX, the primary motivation of employing PCA is to perform preliminary dimensionality reduction on the highly redundant HSI, preventing excessively high computational costs before the features enter the CAFM. The empirical results demonstrate that removing PCA leads

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TABLE XI
COMPARISON OF MODEL PARAMETERS, FLOATING POINT OPERATIONS (FLOPS), AND INFERENCE TIME ON THE AUGSBURG DATASET

Metrics	FusAtNet	S^2ENet	AsyFFNet	ExViT	HCT	MACN	MICF-Net	GCCQTNet	CHNet	MGMNet	RSCNet
Params (M)	36.8236	1.7055	2.0987	1.8807	4.7811	2.0199	1.7694	1.2608	20.7745	6.1251	2.8812
Flops (G)	3.3830	0.2065	0.1789	0.2897	0.0437	0.1672	0.2280	0.0293	0.1609	0.1259	0.3346
Inference time (s)	0.2054	0.2851	0.2468	0.3243	0.7524	0.1917	0.2642	0.4667	0.5582	0.2842	0.3483

to an increase in model parameters across all three datasets. Furthermore, the OA also experiences a slight drop without PCA. This indicates that PCA not only reduces the parameter burden but also mitigates the risk of overfitting caused by raw spectral noise in the fusion stage, thereby achieving an optimal tradeoff between efficiency and accuracy.

6) Effectiveness of KBSM: To demonstrate the superiority of the proposed KBSM over established spectral dimensionality reduction techniques, we conduct a comprehensive comparison against five representative band selection methods: two traditional machine learning approaches, namely improved sparse subspace clustering (ISSC) [59] and orthogonal-projection-based band selection (OPBS) [60]; and three deep learning-based methods, including bottleneck attention module (BAM) [61], self-representation learning with sparse 1-D operational autoencoder (SRL-SOA) [62], and LiDAR-guided cross-attention fusion (LGCAF) [63]. To highlight the effectiveness of different band selection strategies, this experiment is conducted on the Berlin dataset. As shown in Table X, traditional methods (ISSC and OPBS), which are applied to the HSI in the preprocessing stage, yield relatively lower OA. Deep learning methods exhibit improved performance, with the cross-modal guided LGCAF achieving a competitive 77.43%. Nevertheless, our KBSM consistently outperforms all competitors, reaching an OA of 78.19%. This confirms that by dynamically utilizing structural priors from multisource data to guide spectral disambiguation, KBSM is highly effective in selecting the most discriminative and representative bands.

7) Computational Complexity Analysis: As shown in Table XI, RSCNet achieves a highly competitive balance among model complexity, running time, and representational capacity. Specifically, the proposed model contains 2.8812 M parameters, requires 0.3346 GFLOPs, and has an inference time of 0.3483 s on the Augsburg dataset. In terms of parameter count, RSCNet is much more compact compared to heavyweight models such as FusAtNet, CHNet, and MGMNet. Regarding inference time, RSCNet processes multisource data much faster than HCT, CHNet, and GCCQTNet. Although RSCNet incurs slightly higher computational overhead and inference time than lightweight architectures due to the dynamic key band selection directly from the raw HSI data, it successfully avoids excessive parameter redundancy and achieves substantial improvements in classification accuracy. These results verify that our RSCNet ensures state-of-the-art performance while maintaining controllable resource consumption and high execution efficiency, demonstrating strong practical applicability.

8) Effectiveness of Multisource Fusion: To substantiate the necessity of the proposed fusion strategy and investigate the performance improvements brought by multisource complementary information, we compare single-source inputs with our multisource fusion approach. Specifically, the network is trained using only SAR/LiDAR or HSI data, and the OA is reported. As shown in Table XII, the multisource fusion (HSI+SAR/LiDAR) consistently outperforms the best single source (HSI) across all three datasets. Notably, on the Houston2013 dataset, the OA of the fusion strategy achieves a substantial improvement of 3.27% compared to the hyperspectral-only classification. These results clearly indicate that the structural priors provided by SAR/LiDAR are highly complementary to the material-dependent spectral features of HSI. Our RSCNet effectively integrates these distinct physical sources, demonstrating the fundamental advantages and necessity of multisource fusion.

TABLE XII
COMPARISON OF OA (%) BETWEEN SINGLE-SOURCE AND MULTISOURCE FUSION ACROSS THREE DATASETS. THE BOLD DENOTE THE BEST RESULTS

Input	Augsburg	Berlin	Houston2013
SAR / LiDAR Only	86.76	61.25	65.29
HSI Only	89.53	75.63	89.39
HSI + SAR/LiDAR (Ours)	**91.50**	**78.19**	**92.66**

V. CONCLUSION AND FUTURE WORK

In this article, we presented RSCNet for multisource remote sensing data classification, which mainly focuses on two challenges:

The first challenge: *How to reduce the spectral redundancy in hyperspectral data?* To address this, we propose the KBSM, which adaptively identifies task-relevant spectral bands under cross-source guidance, mitigating the limitations of traditional dimensionality reduction and preserving discriminative spectral–spatial information. The second, challenge: *How to enhance the cross-source feature interactions?* To solve this problem, we propose the CAFM, which enables robust feature interaction by performing cross-source attention weighting and local–global contextual refinement. Extensive experiments conducted on three public benchmark datasets demonstrate that our RSCNet achieves superior performance over state-of-the-art methods.

In the future, we plan to further enhance the robustness of the KBSM by improving its stability under complex atmospheric noise and cross-sensor variations. In addition, we plan to extend the proposed framework to support additional data sources, such as multispectral and infrared data, and explore its potential in semi-supervised or self-supervised paradigms to alleviate the reliance on extensive annotated data.

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